Layer 2: Options Volatility Surface

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1 Options Data Ingestion and Handling

Critical technical challenges related to data completeness, consistency, and database management:

1. API Query Logic and Data Completeness A fundamental flaw in the initial API interaction led to incomplete options data, limiting our collection almost entirely to long-term contracts.

- **Root Cause:** Previously, the API call only set a lower bound for the expiration date (`expiration date gte=date str`) without setting an upper limit. Because the Polygon API sorts results by expiration date, our maximum limit of 1000 contracts inadvertently returned only the longest-term contracts (LEAPS), missing all short-term, highly liquid options.
- **Historical Data Fix:** A separate, crucial issue was the omission of the `as of` parameter. The API was returning contracts valid as of today (e.g., 2025-11-04) instead of contracts valid on the specific historical date we were querying (e.g., 2024-01-18).
- **Resolution:** The API call was fixed by adding `as of=day str` to ensure time consistency. Further pagination logic was introduced (or implied) to collect a broader range of expiration dates up to the limit.

2. OSI Code Parsing Inconsistency We encountered an issue with the standard Option Symbol Identifier (OSI) format used for specific historical records.

- **Issue:** While the standard Polygon format is `O:SPY251219C00650000` (six-digit date: YYMMDD), we found an outlier formatted as `O:AMD1240119P00150000`. This seven-digit date section (`1240119`) deviated from the expected six-digit convention.
- **Current Status:** The `parse osi` function was temporarily configured to skip any OSI codes containing a seven-digit date section. This allowed the data ingestion script to continue running without failure, but a more robust handler for this format inconsistency may be required for complete historical coverage.

3. Database Upsert Conflict and Deduplication Direct database interaction required deduplication to prevent conflicts and ensure atomic updates.

- **Issue:** During the batch `INSERT` operation, duplicate `(option name, time entry ts)` pairs within a single batch statement prevented PostgreSQL's `ON CONFLICT DO UPDATE` clause from executing correctly, leading to insertion failure.

- **Resolution:** A deduplication step was added to the `upsert minute rows()` function. This ensures that only records unique on the primary key are included in the batch `INSERT`, allowing the upsert logic to proceed as intended.

Table 1: Option Risk Metrics (The Greeks and Implied Volatility)

Symb.	**Name**	**Measures**	**Intuitive Explanation**
IV	Implied Volatility	Expected future price volatility.	The market's perception of how wild the stock price will swing. Higher IV = more expensive option.
Δ	Delta	Price change for $1 change in stock price.	How much money the option gains/loses for a $1 stock move. Also, the approximate probability of finishing ITM.
Γ	Gamma	Rate of change in Delta (per $1 stock change).	How quickly the option's price sensitivity (Δ) accelerates or decelerates.
Θ	Theta	Rate of change for one-day decrease in time.	The time decay of the option's value. The daily cost of holding the option.
ν	Vega	Price change for 1% change in IV.	How sensitive the option's price is to changes in market anxiety (IV).
ρ	Rho	Price change for 1% change in risk-free rate.	How much the option's value shifts due to interest rate changes.

Currently the data plan for Stock does not cover Real Time or Intraday Option values. This can be solved by puchasing the $250/*mo* Option Package. https://www.alphavantage.co/query?function=HISTORICAL_OPTIONS&symbol=IBM&apikey=demo

This does not make analysis impossible however. End of Day (EOD) Data can still simulate options backtesting.

Table 2: End-of-Day Options Chain Query Structure

Query Date	**Data Structure**
`2025-10-10`	`List[Contract Snapshot]`

Table 3: Contract Listing Structure for Specific Expiration and Strike Range

Index	**Contract ID**	**Type**	**Strike**
0	`IBM251010C00150000`	Call	150.00
1	`IBM251010P00150000`	Put	150.00
2	`IBM251010C00155000`	Call	155.00
3	`IBM251010P00155000`	Put	155.00
...			
20	`IBM251010C00200000`	Call	200.00
21	`IBM251010P00200000`	Put	200.00

Where each of the contract IDs Expand into

Where if queried for a set of days, we can return time series analysis.

Table 4: Structure of a Single Option Contract Snapshot (End-of-Day)

Field Name	Type	Group	Example	Field Name	Type	Group	Example
`contractID`	String	ID	`IBM2509...`	`mark`	Float	Pricing	156.45
`symbol`	String	ID	`IBM`	`bid`	Float	Pricing	154.55
`date`	Date String	Time	`2025-09-15`	`ask`	Float	Pricing	158.35
`expiration`	Date String	Contract	`2025-09-19`	`bid size`	Integer	Liquidity	129
`strike`	Float	Contract	100.00	`ask size`	Integer	Liquidity	129
`type`	String	Contract	`call`	`volume`	Integer	Liquidity	0
`last`	Float	Pricing	144.70	`open interest`	Integer	Liquidity	2
`implied volatility`	Float	Greeks	3.72698	`delta`	Float	Greeks	0.99545
`gamma`	Float	Greeks	0.00013	`theta`	Float	Greeks	-0.17795
`vega`	Float	Greeks	0.00357	`rho`	Float	Greeks	0.01081

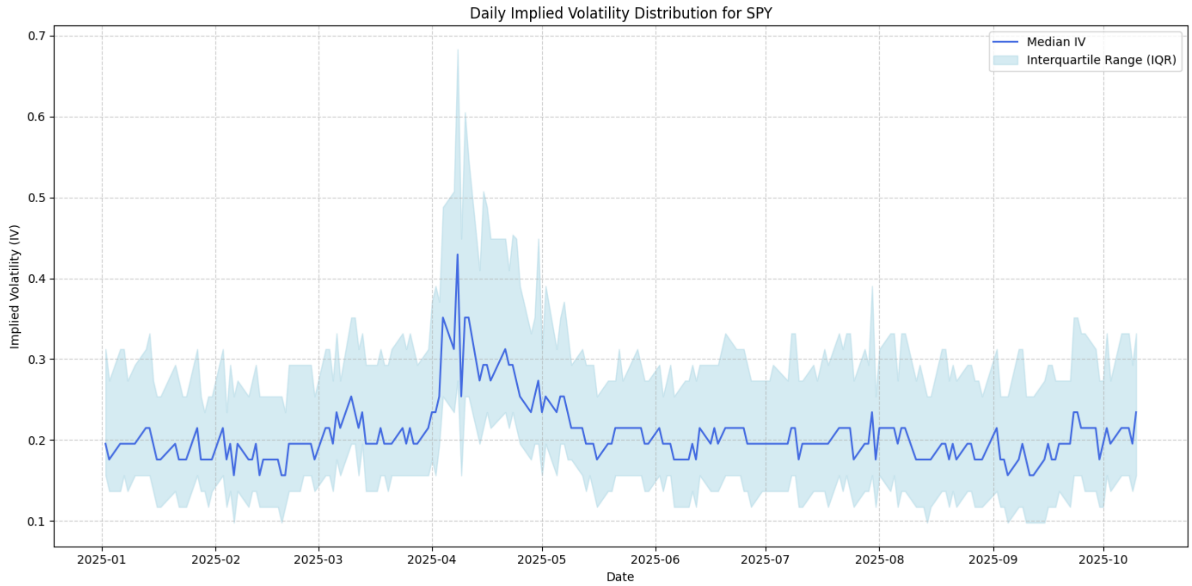

Figure 1: SPY Options Chain Median IV Distribution Over Time

2 Options Data Structure and Identification

An options name (ticker) uniquely identifies the contract and is standardized across major brokerages and data providers.

Sample Data Source Links:

- **Alpha Vantage (Historical Options Sample):** `https://www.alphavantage.co/query?function=HISTORICAL_OPTIONS&symbol=IBM&date=2025-10-10&apikey=...`
- **Polygon.io (Aggregated Options Sample):** `https://api.polygon.io/v2/aggs/ticker/O:IBM251010P00255000/range/1/minute/2025-10-01/2025-10-10?adjusted=true&sort=asc&limit=120&apiKey=...`

To process the options universe, we first standardize individual contracts using the Option Symbol Identifier (OSI) format. This allows for precise decomposition of the asset, expiry, and strike price from a single string identifier.

Example Identifier: `O:IBM251010P00255000`

- **O:** Asset Class (Option Contract).
- **IBM:** Underlying Asset Ticker.

- **251010:** Expiration Date (YY/MM/DD) → 2025/10/10.
- **P:** Contract Type (Put). *Alternately 'C' for Call.*
- **00255000:** Strike Price (scaled by 1000) → $255.00.

It is crucial to note that while we observe the contract at a specific moment (Time-to-Expiry or TTE), the initialization date of that contract varies. A contract observed with 1 Day to Expiry (1 DTE) on 2025/10/09 may have been written by an institutional market maker (e.g., a Hedge Fund or HFT) years prior as a LEAP (Long-Term Equity Anticipation Security).

3 Universe Selection and Anti-Lookahead Filtering

Processing the entire options chain for every tick is computationally infeasible and introduces statistical noise. We dynamically filter the options universe to retain only relevant contracts.

3.1 The Moneyness Filter

We restrict our dataset to contracts within a **±10%** Moneyness Ratio ($M_{\%}$). This creates a fixed-size input window for the machine learning pipeline and filters out deep Out-of-the-Money (OTM) contracts that behave unpredictably.

$$\text{Moneyness}_{\%} = \frac{|S - K|}{S} \leq 0.10 \tag{1}$$

Where S is the anchor price and K is the strike price.

3.2 Anti-Lookahead Anchoring

A critical source of data leakage in financial ML is the use of future knowledge (e.g., End-of-Day close price) to filter data available at the beginning of the day. To eliminate this **Lookahead Bias**, we derive acceptable strike boundaries ($\mathbf{K}_{\min}$ and $\mathbf{K}_{\max}$) using the Start-of-Day (SOD) Open Price.

1. **Anchor (S):** The underlying asset price at Market Open (e.g., 13:30 UTC).
2. **Boundaries:**
$$K_{\min} = S \times 0.90 \quad \text{and} \quad K_{\max} = S \times 1.10$$

4 Data Dimensionality and Surface Visualization

The filtered data is constructed into a multi-dimensional tensor representing the volatility surface.

- **X-Axis (Strike):** The raw Strike Price (K). Because the filtering anchor (S) changes daily, the absolute range of this axis "jumps" day-to-day, but always represents the ±10% window relative to SOD.
- **Y-Axis (Expiry):** A floating 30-day window of maturity. For example, if the observation date is 9/01, the axis spans contracts expiring between 9/02 and 10/01.
- **Z-Axis (Target):** Represents Implied Volatility (Task 1) or Option Price (Task 2).

The resulting visualization consists of two interleaved surfaces (Put and Call) that converge at the underlying asset price.

5 Theoretical Framework: BSM and Volatility Skew

To understand the relationship between the Z-axis (Price/IV) and the X-axis (Strike), we utilize the Black-Scholes-Merton (BSM) model. While BSM assumes a constant volatility (σ), market

Figure 2: BSM Call Option Pricing Formula Derman et al. [1996]

$$C = S_0 N(d_1) - K e^{-rT} N(d_2) \tag{2}$$

Where auxiliary terms d_1 and d_2 are:

$$d_1 = \frac{\ln\left(\frac{S_0}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) T}{\sigma\sqrt{T}} \tag{3}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{4}$$

observations reveal a "Volatility Skew," where implied volatility varies across strike prices.

Calls								Puts					
Exp. Date	Last	Change	Bid	Ask	Volume	Open Int.	Strike	Last	Change	Bid	Ask	Volume	Open Int.
October 17, 2025													
Oct 17	22.34	-0.43 ▼	22.65	23.40	1493	12484	225.00	0.07	-0.03 ▼	0.07	0.09	754	12628
Oct 17	21.40	+0.04 ▲	19.85	21.10	29	1202	227.50	0.09	-0.03 ▼	0.09	0.10	428	2267
Oct 17	17.55	-0.40 ▼	17.70	18.40	285	19155	230.00	0.11	-0.04 ▼	0.11	0.12	1140	12434
Oct 17	15.40	-0.67 ▼	15.30	16.20	11	265	232.50	0.14	-0.07 ▼	0.14	0.16	1930	3034
Oct 17	13.25	-0.18 ▼	12.95	13.50	1044	14877	235.00	0.21	-0.08 ▼	0.21	0.23	6842	11338
Oct 17	10.40	-1.01 ▼	10.45	11.10	237	254	237.50	0.33	-0.09 ▼	0.32	0.35	6736	5264
Oct 17	8.40	-0.10 ▼	8.40	8.65	3278	26795	240.00	0.53	-0.13 ▼	0.54	0.56	14237	13073
Oct 17	6.45	+0.10 ▲	6.25	6.45	825	699	242.50	0.91	-0.14 ▼	0.89	0.93	9219	5426
Oct 17	4.50	+0.05 ▲	4.35	4.55	5300	15514	245.00	1.50	-0.20 ▼	1.49	1.53	22299	14552
Oct 17	2.95	--	2.84	2.91	21361	7399	247.50	2.42	-0.26 ▼	2.40	2.47	9992	8003
Oct 17	1.66	-0.14 ▼	1.66	1.71	37358	36506	250.00	3.75	-0.24 ▼	3.65	3.85	3185	10209
Oct 17	0.88	-0.14 ▼	0.88	0.91	20198	7294	252.50	5.92	+0.57 ▲	5.35	5.55	674	5173
Oct 17	0.45	-0.11 ▼	0.43	0.45	37269	27740	255.00	8.04	+0.09 ▲	7.30	7.70	526	6935
Oct 17	0.21	-0.09 ▼	0.20	0.21	17841	19834	257.50	10.29	+0.20 ▲	9.55	10.00	559	3552
Oct 17	0.09	-0.07 ▼	0.09	0.10	24495	39171	260.00	12.65	+0.15 ▲	12.00	12.50	169	3298
Oct 17	0.05	-0.04 ▼	0.04	0.05	15180	26650	262.50	14.29	+0.04 ▲	14.25	15.00	1714	1938
Oct 17	0.03	-0.03 ▼	0.02	0.03	4601	23159	265.00	16.50	-1.70 ▼	16.65	17.55	104	107
Oct 17	0.01	-0.03 ▼	0.01	0.02	3427	7980	267.50	19.27	+0.20 ▲	19.15	20.05	20	4
Oct 17	0.02	-0.02 ▼	0.01	0.02	2720	20287	270.00	21.72	+0.07 ▲	21.80	22.65	52	8
Oct 17	0.01	-0.01 ▼	--	0.01	223	3732	272.50	24.32	+1.27 ▲	24.20	25.05	10	2

Figure 3: AAPL options chain (Oct 17, 2025 Expiry, ≈ 3 DTE

5.1 Case Study: Anatomy of an Options Chain (AAPL)

To validate the existence of Volatility Skew and justify separating Put and Call surfaces, we analyze a snapshot of the AAPL options chain (Oct 17, 2025 Expiry, ≈ 3 DTE).

Table 5: AAPL Context Data (Snapshot)

Metric	Value
Underlying Asset	AAPL (Dividend Paying)
52-wk Range	169.21 - 260.09
Current Price	≈ \$247.50

By observing the chain, we derive several critical market dynamics that our model must capture:

1. **Put-Call Parity Dislocation (Skew):** Puts are consistently more expensive than equivalent moneyness Calls by approximately 15%.

   - *Why?* Institutional demand for downside protection (portfolio insurance) drives up the premium for OTM Puts. This confirms that the Volatility Surface is not symmetric. Bollen and Whaley [2002]

2. **Bearish Outlook Indicators:** In this specific snapshot, Put prices gained value (Avg $+0.20) while Call prices lost value (Avg -0.50$). This suggests immediate bearish sentiment on the underlying asset.

3. **Liquidity Constraints:** The Bid-Ask spread widens significantly as we move deeper Out-of-the-Money (OTM).

   - *Implication:* Our $\pm 10\%$ filter is not just for data reduction, but for avoiding "Liquidity Canyons" where wide spreads introduce noisy pricing data.

4. **Early Exercise Premium (EEP):** Although AAPL is a dividend-paying stock, the short Time-to-Expiry (3 DTE) means the discount factor e^{-rT} is negligible. We observe minimal divergence between American and European pricing in this specific window.

5.2 Local Volatility Surface

Given the observed Skew, a constant σ is insufficient. We must account for the local volatility σ_{local}, which changes as the strike approaches the underlying price Derman et al. [1996].

$$\sigma^2_{\text{local}}(K, T) = \frac{\frac{\partial C}{\partial T} + rK\frac{\partial C}{\partial K}}{\frac{1}{2}K^2\frac{\partial^2 C}{\partial K^2}} \tag{5}$$

Our system's goal is to predict the **Local Volatility** Derman et al. [1996], an unbiased volatility surface that accounts for these skew dynamics, allowing us to solve for the true market price of the contract.

6 Feature Extraction and Volatility Surface Modeling

The feature extraction pipeline transforms raw option prices into a parametric representation of the volatility surface. This reduction is critical for the machine learning model, as predicting a dense matrix of option prices directly is computationally intractable and prone to overfitting. We utilize the SABR model framework, as explored in Thorin [2020], to capture the volatility smile and skew dynamics.

6.1 Prediction Targets

The ultimate goal is to predict the parameters that describe the future volatility surface at time $t + 1$. The machine learning model targets a tuple of four key volatility indicators:

- LV_{t+1}: The **Local Volatility**, a VIX-style measure representing the overall market volatility level.
- α_{t+1}: The SABR **ATM Volatility** parameter for the key expiry.
- ρ_{t+1}: The SABR **Correlation** parameter (Skew) for the key expiry.

- ν_{t+1}: The SABR **Vol-of-Vol** parameter (Smile curvature) for the key expiry.

Secondary targets may include the underlying forward price F_{t+1}, risk-free rate r_{t+1}, and dividend yield q_{t+1}.

Surface Reconstruction Once these parameters are predicted, the volatility surface $\sigma_{\text{IV}}(K, T)$ for a specific strike K and expiry T can be analytically reconstructed using Hagan's formula:

$$\sigma_{\text{IV}}(K, T) \approx \text{SABR}(F_{t+1}, K, \alpha_{t+1}, \beta, \rho_{t+1}, \nu_{t+1}) \tag{6}$$

This approach focuses on predicting the dynamics of the *parameters* driving the surface, rather than the surface itself.

6.2 Methodology: Computing Volatility Tuples

We process minute-by-minute market data containing the timestamp t, underlying price S_t, and the full chain of call and put option prices. We assume a risk-free rate $r \approx 4.5\%$ and a continuous dividend yield $q \approx 0.40\%$.

The extraction process consists of two distinct calibration tasks performed at each timestep.

6.2.1 Task 1: LV (VIX-Style) Calculation

The Local Volatility (LV) provides a model-free metric of variance. We adopt the CBOE VIX methodology, which aggregates weighted prices of Out-of-the-Money (OTM) options.

For a near-term expiry T, the variance σ^2 is approximated by summing across all valid strikes K_i:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left[\frac{F}{K_0} - 1 \right]^2 \tag{7}$$

Where:

- $Q(K_i)$: The midpoint price of the OTM option at strike K_i.
- F: The forward index level.
- K_0: The first strike below the forward index level F.

This process is repeated for two maturities (T_1, T_2) bracketing a 30-day target horizon. The final LV_t is obtained by linear interpolation of the variances $\sigma^2_{T_1}$ and $\sigma^2_{T_2}$ followed by a square root.

6.2.2 Task 2: SABR Model Calibration

While LV captures the *level* of volatility, the SABR model captures the *shape* (skew and smile). We calibrate the Stochastic Alpha, Beta, Rho (SABR) model parameters to a key representative expiry T_{key} (e.g., closest to 30 days).

Step A: Invert Black-Scholes

First, we convert market option prices P_{mkt} into market implied volatilities $\sigma^{\text{mkt}}_{\text{IV}}$ by numerically inverting the Black-Scholes pricing formula:

$$\sigma^{\text{mkt}}_{\text{IV}}(K) = \text{BS}^{-1}(P_{\text{mkt}}(K), S_t, K, T_{\text{key}}, r, q) \tag{8}$$

Step B: Optimize SABR Parameters

We fix the elasticity parameter β (typically $\beta = 0.5$ or $\beta = 1.0$ for equities) to ensure calibration stability Hagan et al. [2002]. We then solve the non-linear least squares optimization problem to find the optimal triplet $(\alpha_t, \rho_t, \nu_t)$:

$$\min_{\alpha_t,\rho_t,\nu_t} \sum_K \left[\sigma_{\text{IV}}^{\text{SABR}}(F_t, K; \alpha_t, \beta, \rho_t, \nu_t) - \sigma_{\text{IV}}^{\text{mkt}}(K)\right]^2 \tag{9}$$

Where $\sigma_{\text{IV}}^{\text{SABR}}$ is Hagan's asymptotic approximation formula Hagan et al. [2002]. This calibration compresses the entire volatility smile of the key expiry into three observable state variables for the ML pipeline.

7 Barone-Adesi-Whaley Price to Implied Volatility (σ_{IV})

The Implied Volatility (σ_{IV}) for an American option is derived using the Barone-Adesi and Whaley (BAW) approximation Barone-Adesi and Whaley [1987]. This involves numerically solving a non-linear equation that equates the theoretical BAW price, $A_{\text{BAW}}(\sigma)$, to the observed market price, A_{market}.

The σ_{IV} is the root σ that satisfies the residual function:

$$f(\sigma; S, K, T, r, q) = A_{\text{market}} - A_{\text{BAW}}(\sigma) = 0 \tag{10}$$

7.1 I. The BAW Price Formula $A_{\text{BAW}}(\sigma)$

The BAW approximation decomposes the American option price into the standard European Black-Scholes price (P_{BS}) plus an Early Exercise Premium (E):

$$A_{\text{BAW}}(\sigma) = P_{\text{BS}}(\sigma) + E(\sigma) \tag{11}$$

7.1.1 A. European Price (P_{BS})

Using the standard Black-Scholes formulation with implied volatility σ:

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}} \tag{12}$$

$$d_2 = d_1 - \sigma\sqrt{T} \tag{13}$$

The European prices for Puts and Calls are:

$$P_{\text{BS}}^{\text{Put}} = Ke^{-rT}\Phi(-d_2) - Se^{-qT}\Phi(-d_1) \tag{14}$$

$$P_{\text{BS}}^{\text{Call}} = Se^{-qT}\Phi(d_1) - Ke^{-rT}\Phi(d_2) \tag{15}$$

7.1.2 B. Early Exercise Premium (E)

The early exercise premium E is non-zero only when the asset price S crosses a critical threshold S^*.

$$E(\sigma) = \begin{cases} A_2\left(\frac{S}{S^*}\right)^{m_2} & \text{if } S < S^* \text{ (Put)} \\ A_1\left(\frac{S}{S^*}\right)^{m_1} & \text{if } S > S^* \text{ (Call)} \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

Note: The exponent for the Put case (m_2) is typically negative, while m_1 for the Call is positive.

7.2 II. BAW Parameters and Critical Price (S^*)

The computational complexity arises from determining the Critical Stock Price (S^*), the boundary where early exercise becomes optimal. S^* is itself a function of σ.

7.2.1 A. Put Parameters (for σ_{Put})

1. **Exponent m_2 (Negative Root):**

$$m_2 = \frac{-(N-1) - \sqrt{(N-1)^2 + 4M}}{2} \tag{17}$$

   Where $N = \frac{2(r-q)}{\sigma^2}$ and $M = \frac{2r}{\sigma^2}$.

2. **Critical Price S^*:** The solution to the implicit equation:

$$K - S^* = P_{\text{BS}}^{\text{Put}}(S^*) - \left(1 - e^{-qT}\Phi(-d_1(S^*))\right)\frac{S^*}{m_2} \tag{18}$$

3. **Coefficient A_2:**

$$A_2 = \frac{S^*}{m_2}\left(1 - e^{-qT}\Phi(-d_1(S^*))\right) \tag{19}$$

7.2.2 B. Call Parameters (for σ_{Call})

1. **Exponent m_1 (Positive Root):**

$$m_1 = \frac{-(N-1) + \sqrt{(N-1)^2 + 4M}}{2} \tag{20}$$

2. **Critical Price S^*:** The solution to the implicit equation:

$$S^* - K = P_{\text{BS}}^{\text{Call}}(S^*) + \left(1 - e^{-qT}\Phi(d_1(S^*))\right)\frac{S^*}{m_1} \tag{21}$$

3. **Coefficient A_1:**

$$A_1 = \frac{S^*}{m_1}\left(1 - e^{-qT}\Phi(d_1(S^*))\right) \tag{22}$$

7.3 III. Handling Missing Option Types

Implied volatility calculation via inversion requires the specific option's market price. Unlike European options, **Put-Call Parity does not hold strictly for American options** due to the early exercise premium. Therefore, IV cannot be reliably inferred for a missing Call using only the Put price (and vice versa).

- If only $A_{\text{market}}^{\text{Put}}$ is available: Solve $A_{\text{market}}^{\text{Put}} - A_{\text{BAW}}^{\text{Put}}(\sigma) = 0$.
- If only $A_{\text{market}}^{\text{Call}}$ is available: Solve $A_{\text{market}}^{\text{Call}} - A_{\text{BAW}}^{\text{Call}}(\sigma) = 0$.

7.4 IV. Solver Performance: Brentq vs. Newton

Calculating σ_{IV} involves nested root-finding: an inner loop for S^* and an outer loop for σ.

- **Performance:** The `brentq` method (Brent's method) demonstrated superior performance ($\approx$ 119.8s vs $\approx$ 146.8s for Newton).
- **Robustness:** The BAW formula is highly non-linear. Brent's method, which combines bisection, secant, and inverse quadratic interpolation, avoids the divergence issues common with Newton-Raphson when the derivative approaches zero.
- **Constraint:** `brentq` requires a valid bracketing interval $[a, b]$ where signs are opposite, necessitating an adaptive bracketing routine.

8 Collapsing Put/Call IV Matrices

We employ a volume-weighted average approach to merge separate Put and Call IV surfaces into a single, robust volatility surface. This is defined as:

$$IV_{\text{Final}} = \frac{IV_{\text{Put}} \cdot V_{\text{Put}} + IV_{\text{Call}} \cdot V_{\text{Call}}}{V_{\text{Total}}} \tag{23}$$

where V represents the trading volume (liquidity weight) at a specific strike and expiry.

8.1 Volume Weighting vs. Hard Selection

Volume weighting is theoretically superior to "Hard Splice" methods (e.g., exclusively using OTM Puts and OTM Calls) for American options, as it preserves surface continuity and utilizes the full spectrum of liquidity information Jiang and Tian [2005].

Table 6: Comparison of Surface Collapsing Methodologies

Feature	Volume Weighting (Proposed)	Hard Splice (Baseline)
Data Integrity	**Inclusive.** Uses all market data. Even if Call IV is noisy, including it with a low weight ($w < 0.1$) anchors the surface to the aggregate market view Hansen and Lunde [2006].	**Lossy.** Discards potential price discovery or arbitrage signals contained in the illiquid/ITM side.
Continuity	**Smooth.** Ensures differentiable transitions across the ATM region where liquidity naturally shifts from Calls to Puts Fengler [2009].	**Discontinuous.** Creates artificial "kinks" or jumps at the arbitrary strike boundary where the source switches.
Liquidity Alignment	**Dynamic.** Automatically prioritizes the most liquid contracts (typically OTM) without hard-coded rules.	**Static.** Binary assumption that ITM options have zero value for volatility estimation, which is factually incorrect.
Robustness	**Dampened.** Outliers in one chain are diluted by the presence of the other, preventing single-quote errors from dominating.	**Fragile.** If the selected side contains a quote error, the final surface adopts that error directly.

9 Risk-Free Rate (r)

The risk-free rate (r) represents the theoretical return of an investment with zero risk of financial loss. It is used primarily to discount the expected future payoff of the option back to the present value.

9.1 The Modern Standard: SOFR

Historically, the London Interbank Offered Rate (LIBOR) was the standard proxy for r. However, following the cessation of LIBOR, the industry has shifted to the **Secured Overnight Financing Rate (SOFR)** Alternative Reference Rates Committee (ARRC) [2023].

- **Proxy:** The standard risk-free curve is derived from the SOFR Overnight Index Swap (OIS) market.
- **Nature:** Unlike LIBOR (which was unsecured), SOFR is fully collateralized by U.S. Treasuries, making it a truer representation of a "risk-free" rate Copeland et al. [2024].

9.2 Practical Substitute: Federal Funds Rate (FFR)

In the absence of a clean SOFR OIS curve, the **Effective Federal Funds Rate (EFFR)** may be used as a substitute, subject to specific adjustments.

9.2.1 Rationale for Substitution

Both SOFR and EFFR track the shortest, safest rates in the U.S. financial system. Since the Federal Reserve sets the target range for FFR, and SOFR tracks closely with effective money market rates, FFR serves as a reasonable short-term anchor for the risk-free environment in simplified models.

9.2.2 Necessary Adjustments

Using the raw target rate is insufficient for accurate pricing. Three adjustments are required:

1. **Use Effective Data:** Do not use the Fed target range. Use the *Effective* Federal Funds Rate (EFFR), which is the volume-weighted median of actual overnight transactions Duffie [2016].
2. **Term Structure:** The EFFR is an overnight rate. For options with maturity $T > 1$ day, you must estimate the market's expectation of the average EFFR over the life of the option.
   - *Best Practice:* Utilize FFR Futures prices to infer the implied term rate for maturity T, rather than assuming a flat current rate. Hull [2018].
3. **Continuous Compounding:** Market rates (like EFFR) are quoted as annualized simple interest (Day count convention usually ACT/360). The Black-Scholes model requires a continuously compounded rate (r_{cc}). The conversion is:

$$r_{cc} = m \ln\left(1 + \frac{r_{simple}}{m}\right) \tag{24}$$

   Where m is the compounding frequency (typically $m = 360$ for U.S. money markets). Hull [2018].

9.3 Modern Multi-Curve Framework

In advanced post-crisis valuation, the definition of the risk-free rate r is bifurcated into two distinct functions to address credit and liquidity risks separately Mercurio [2009] Bianchetti [2012]:

- **Discounting (P_V):** Use a collateralized rate (e.g., SOFR or OIS) to discount cash flows. This reflects the risk-free nature of variation margin posting.
- **Forward Pricing (F):** Use a funding cost rate (or cost-of-carry rate) to project the forward price of the underlying asset, reflecting the specific funding constraints of the institution.

10 Contango and Backwardation in American Option Pricing

The relationship between the risk-free rate (r) and the continuous dividend yield (q) determines the shape of the forward curve, which fundamentally drives the early exercise strategy for American options.

- Let S be the current stock price, K be the strike price.
- The non-dividend paying forward price is defined as $F = Se^{(r-q)T}$.

10.1 Contango ($r > q$)

Definition: Contango occurs when the cost of holding the underlying asset (the funding cost/risk-free rate, r) exceeds the benefit received from holding it (the dividend yield, q). Consequently, the forward price trades at a premium to the spot price ($F > S$) Hull [2018]

Put Option Early Exercise Decision:

- **Financial Intuition:** "Cash is king; sell the asset early."
- **Mechanism (American Put):** A high future price (F) relative to spot (S) erodes the value of a Put option over time. The incentives align for early exercise.
- **Formal Reason:** In a Contango environment, it becomes optimal to exercise deep In-The-Money (ITM) Puts early Merton [1973].
  1. **Capital Efficiency:** By exercising, the holder receives the strike price K immediately at time t.
  2. **Interest Arbitrage:** This cash proceeds can be immediately reinvested at the higher risk-free rate (r).
  3. **Opportunity Cost:** Delaying exercise ties up capital in the asset which yields a lower net return (q), forfeiting the risk-free interest income.
- **Condition:** Early exercise is optimal when the intrinsic value ($K - S$) plus the interest earned on K exceeds the option's remaining time value and volatility potential.

10.2 Backwardation ($r < q$)

Definition: Backwardation occurs when the benefit of holding the underlying asset (dividend yield, q) exceeds the cost of financing it (r). Consequently, the forward price trades at a discount to the spot price ($F < S$).

Put Option Early Exercise Decision:

- **Financial Intuition:** "Don't sell the golden goose; hold the stock."
- **Mechanism (American Put):** The high dividend yield acts as a strong disincentive against early exercise.
- **Formal Reason:** It is rarely optimal to exercise an American Put early in deep backwardation Hull [2018].
  1. **Delivery Penalty:** Exercising the Put requires delivering the stock to the writer.
  2. **Dividend Forfeiture:** By delivering the stock, the holder instantly forfeits all future dividend payouts (q), which are the dominant source of value in this regime.
  3. **Strategic Holding:** The value of the dividend stream outweighs the interest that could be earned on the strike price K. The rational agent holds the option to retain exposure to the stock's yield.

- **Condition:** Early exercise is generally suboptimal unless a discrete dividend payment is imminent and sufficiently large to disrupt this equilibrium.

11 Robust Descent Seed eSABR

11.1 Heuristic Initialization

To ensure the convergence of the Extended SABR (eSABR) optimization, we construct a robust initial guess vector $\mathbf{x}_0$. This seed is derived heuristically to prevent the optimizer from getting trapped in local minima common in the high-dimensional volatility surface space Hagan et al. [2002].

Table 7: Extended SABR (eSABR) Initial Guess ($\mathbf{x}_0$) Construction

Idx	Param	Heuristic Derivation (Robust Estimation Logic)
0	α_S	**Short-term Level:** The ATM volatility of the first liquid expiry slice T_1. $\alpha_S = \sigma_{\text{ATM}}(T_1)$
1	α_L	**Long-term Level:** The median ATM volatility of the latter half of the term structure. Using the median rejects artifacts often found in illiquid back-month contracts. $\alpha_L = \text{Median}\left(\{\sigma_{\text{ATM}}(t) \mid t \in [T_{\text{mid}}, T_{\text{last}}]\}\right)$
2	k	**Decay Rate:** Calculated to explicitly match the volatility drop between the first (T_1) and second (T_2) layers, capturing the initial steepness of the term structure. $k \approx -\frac{1}{T_2} \ln\left(\frac{\sigma_{\text{ATM}}(T_2)-\alpha_L}{\alpha_S-\alpha_L}\right)$
3	ν_S	**Short-term Curvature:** Approximation based on the smile slope of T_1, utilizing an 80% moneyness cut to exclude deep OTM outliers West [2005]. $\nu_S \approx \frac{2\cdot\lvert\text{Slope}(T_1)\rvert}{\alpha_S}$
4	ν_L	**Fixed Asymptote: Fixed to 0.1.** This overrides the heuristic calculation to regularize the noisy long-end search space, preventing the "vol-of-vol" from exploding where liquidity is sparse. $\mathbf{x}_0[4] = 0.1$
5	ρ	**Correlation:** Determined by the sign of the skew slope at T_1. $\rho \approx \begin{cases} -0.6 & \text{if Slope}(T_1) < 0 \\ +0.1 & \text{otherwise} \end{cases}$

Initial Guess Vector $\mathbf{x}_0$:

$$\mathbf{x}_0 = [\alpha_S, \alpha_L, k, \nu_S, \nu_L, \rho]^T \tag{25}$$

11.2 Term Structure Parameter Decay Function

The volatility surface parameters are modeled using a hybrid decay function anchored at the first observable time to expiry ($T_{\min}$). This approach ensures stability (α_S is tied to the most liquid data layer) while using physics-aligned decay forms: a Hyperbolic decay for the sharp short-term volatility level (α), and an Exponential decay for the smoother curvature (ν) Gatheral [2012].

11.2.1 1. Time Shift and Decay Functions

The effective decay time (Δt) is calculated relative to the shortest expiry in the dataset:

$$\Delta t = \max(T - T_{\min}, 0) \tag{26}$$

The specific decay kernels are defined as:

- **Alpha Decay (Inverse Square Root / Hyperbolic):** Used for the level parameter (α). This function naturally models the steep drop (the "cliff") in volatility often observed near the front month:
$$D_\alpha(\Delta t, k) = \frac{1}{1 + k\sqrt{\Delta t}} \tag{27}$$

- **Nu Decay (Exponential):** Used for the Vol-of-Vol parameter (ν). This models the slower, smoother mean-reversion of market curvature:
$$D_\nu(\Delta t, k) = e^{-k\Delta t} \tag{28}$$

11.2.2 2. Parameter Interpolation and Optimization

The final parameter values at time T are interpolated between the short-term estimates and the long-term asymptotic floors:

$$\alpha(T) = \alpha_L + (\alpha_S - \alpha_L)D_\alpha(\Delta t, k) \tag{29}$$
$$\nu(T) = \nu_L + (\nu_S - \nu_L)D_\nu(\Delta t, k) \tag{30}$$

Optimization Objective:

$$\mathbf{x}^* = \underset{\mathbf{x}}{\operatorname{argmin}} \left(\frac{1}{W} \sum_{i=1}^{N} w_i \cdot (\sigma_{\text{model}}(\mathbf{x}, K_i, T_i) - \sigma_{\text{mkt},i})^2 \right) \tag{31}$$

Where weights w_i are set to prioritize the liquid front-month (T_1) to anchor the fit (e.g., $w_{T_1} = 2.0$).

12 Theoretical Sparse Time Field Oversampling

12.1 Forward Fill IV One Step: Theoretical Justification

The practice of performing a one-step forward fill of Implied Volatility (IV) serves as an essential **Engineering Prior** to manage the sparsity of real-time option data.

12.1.1 Does this break the Martingale Property?

No, provided the forward fill is applied to the **IV (σ)** and not the Option Price (P).

1. **Price Fill ($\mathbf{P_{t-1} \to P_t}$):** This introduces **Static Arbitrage**. If the Spot (S_t) moves but the Option Price (P) is held constant, the implied volatility shifts non-linearly and often creates arbitrageable violations of the Black-Scholes boundaries.

2. **IV Fill ($\boldsymbol{\sigma_{t-1} \to \sigma_t}$):** This relies on the **Sticky Strike** assumption Derman [1999]. This asserts that the market's view on volatility for a fixed strike has not changed in the brief interval Δt. Using σ_{t-1} with the *current* Spot S_t to recalculate a synthetic price is mathematically consistent and preserves the arbitrage-free condition of the instantaneous price relationship.

12.2 Variance Marking and Data Oversample

When feeding data into an Extended Kalman Filter (EKF), the uncertainty of each observation is defined by the **Measurement Noise Covariance Matrix (R)**. If a fabricated measurement (σ_{t-1}) is treated the same as a fresh trade (σ_t), the filter commits **Data Oversample** (overweighting stale information).

12.2.1 Implementation: The Dynamic R Matrix

To prevent filter divergence, we dynamically adjust the variance term in R based on a data quality mask Simon [2006]:

1. **Real Trade:** Assign low variance in R (High Confidence).
2. **Forward Filled:** Assign high variance in R (Low Confidence).

This effectively forces the EKF to rely more on its internal **Model Prediction (Dynamics)** when data is stale, and "snap" to the measurement only when fresh market data arrives.

12.3 Alternative: The "Skip" Method

If the data is extremely sparse, an alternative is to use the EKF's native capability to handle missing data:

- **Prediction Step ($\mathbf{x}_{k|k-1}$):** Always run. This evolves the physical state (decay, mean reversion).
- **Update Step ($\mathbf{x}_{k|k}$): Skip** the update for dimensions where data is missing.

Mathematically, skipping the update is equivalent to setting the measurement variance for that point in R to infinity. However, the **Dynamic R Matrix approach** is preferred for GPU implementation as it maintains dense data tensors, allowing for efficient vectorization while still applying the necessary regularization.

13 Forcing the Pre-Kalman Fit

13.1 Problem Statement: The Ill-Posed Inverse Problem

During the calibration of the parametric Extended SABR (eSABR) curve, we identified fundamental instabilities in the static initialization phase. The optimization process exhibited sensitivities characteristic of an ill-posed inverse problem **?**, specifically:

- **High-Leverage Overfitting:** The initialization logic (based on the last expiry layer) was highly susceptible to data sparsity. A single erroneous observation in the furthest expiry (e.g., IV 0.47 vs. valid 0.42) would skew the entire initial guess, causing the optimizer to trap in local minima.
- **Jagged Minima Artifacts:** The fitted curve frequently produced minima significantly lower than the observed market data (e.g., fitted 0.54 vs. observed 0.59) with inflated Vol-of-Vol (ν_S) to compensate. This violated the smoothness constraint of the volatility smile Gatheral [2012].
- **Stubborn Term Structure Decay:** The standard exponential decay model failed to capture the sharp "cliff" often observed between the first (T_1) and second (T_2) expiries (e.g., a drop from 0.76 to 0.50). The solver minimized global error by "smearing" this drop, resulting in drastic underestimation of the first layer and overestimation of the second.

13.2 Empirical Diagnosis: Global Least Squares Bias

The root cause was identified as **Global Least Squares Bias**. The unweighted optimizer treated all data points equally, sacrificing the precision of the high-signal front months to accommodate low-signal, noisy back months. Additionally, the theoretical exponential decay e^{-kt} lacks the geometric steepness required to model short-term event volatility without forcing the decay parameter k to numerically unstable levels ($k > 50$).

13.3 The Engineered Solution

To force a stable pre-Kalman fit, we implemented a multi-stage regularization strategy that combines robust statistics with physics-aligned decay functions.

13.3.1 1. Robust Descent Seed (Median Initialization)

We refactored the initialization logic to use robust statistics. Instead of relying on the last expiry layer, the long-term parameters (α_L, ν_L) are estimated using the **Median** of the latter half of the term structure. This filters out sparse artifacts and provides a stable center of gravity for the descent Huber [1964].

13.3.2 2. Hyperbolic Decay (Physics-Aligned Modeling)

We replaced the Exponential Decay model for the α (Level) parameter with a **Shifted Hyperbolic Decay** (Inverse Square Root). This aligns with the empirical "Square Root of Time" rule for volatility variance, allowing the model to capture sharp initial drops naturally without parameter explosion J.P. Morgan/Reuters [1996].

$$\alpha(t) = \alpha_L + (\alpha_S - \alpha_L) \cdot \frac{1}{1 + k\sqrt{t}} \tag{32}$$

13.3.3 3. Weighted Anchoring (Soft Constraint)

We implemented a **Weighted Least Squares** objective. The liquid first expiry layer is assigned a higher weight ($w = 2.0$) compared to the rest of the term structure. This acts as a "Soft Anchor," forcing the optimizer to prioritize the accuracy of the front-month fit while maintaining global coherence Nelson and Siegel [1987].

14 SSVI (Surface Stochastic Volatility Inspired)

14.1 Motivation and Design Goals

The goal of the SSVI module is not to "win a beauty contest" against local volatility models like SABR, but to engineer an arbitrage-free volatility surface that satisfies three engineering constraints:

1. **Fit:** Fits the cross-sectional Implied Volatility (IV) data with low loss on liquid, near-the-money points.

2. **Structure:** Admits a simple parametric term structure in time to expiry, preventing calendar arbitrage Gatheral [2012].

3. **Stability:** Is numerically stable enough to serve as a downstream feature generator and input to a temporal smoother (e.g., Kalman Filter).

We retain the theoretical structure of SSVI but explicitly regularize both the term-structure parameterization and the cross-sectional loss. Furthermore, we calibrate calls and puts separately at each minute to capture microstructure noise before merging.

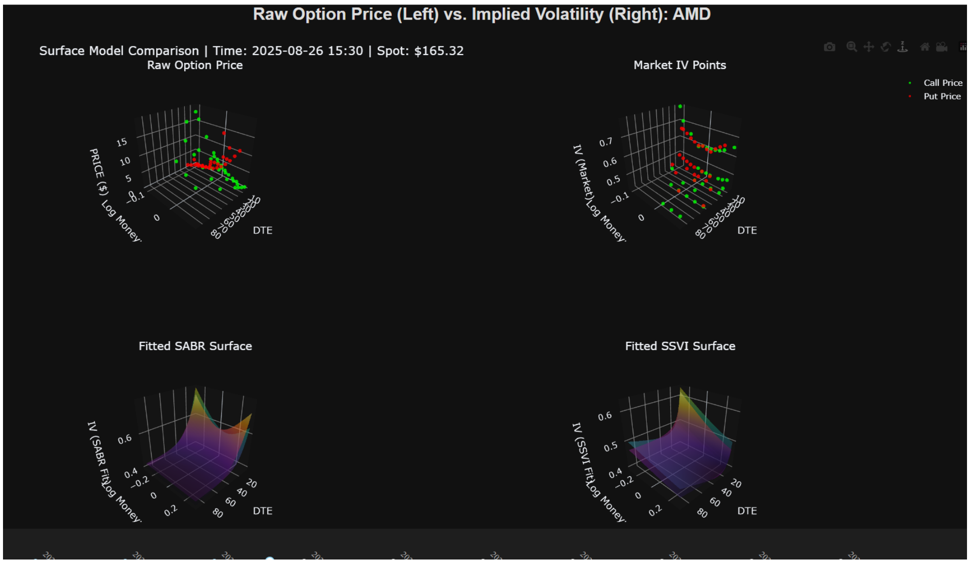

Figure 4: Optimal eSABR fit demonstrating the corrected term structure decay and robust smile capture achieved via Hyperbolic Decay and Weighted Anchoring.

14.2 Parametric SSVI Specification

We employ a single global SSVI parameterization with a quadratic term structure for both the total variance level $\theta(T)$ and the smile intensity $\lambda(T)$. At each timestamp t, the state vector $\mathbf{c}$ consists of 8 parameters:

$$\mathbf{c} = \big(\rho,\ \gamma,\ \lambda_0, \lambda_1, \lambda_2,\ c_0, c_1, c_2\big) \in \mathbb{R}^8$$

where:

- ρ: Global skew correlation parameter (constrained to $[-0.95, 0.95]$).
- γ: Controls the curvature of the SSVI smile (constrained to $\gamma \in [0.3, 1.2]$).
- $(\lambda_0, \lambda_1, \lambda_2)$: Coefficients defining the quadratic term structure for smile intensity.
- (c_0, c_1, c_2): Coefficients defining the quadratic term structure for ATM total variance.

Given time to expiry T (in years), the term structures are defined as:

$$\theta(T) = c_0 + c_1 T + c_2 T^2 \tag{33}$$

$$\lambda(T) = \lambda_0 + \lambda_1 T + \lambda_2 T^2 \tag{34}$$

In implementation, both functions are clipped to enforce positivity ($\theta(T) \geq \theta_{\min} \approx 10^{-4}$, $\lambda(T) \geq \lambda_{\min} \approx 10^{-6}$).

For log-moneyness $k = \log(K/F)$, the SSVI total variance $w(k,T)$ is given by the standard form Gatheral [2012]:

$$w(k,T) = \frac{\theta(T)}{2}\left(1 + \rho\phi_T k + \sqrt{(\phi_T k + \rho)^2 + (1-\rho^2)}\right) \tag{35}$$

where $\phi_T = \frac{\lambda(T)}{\theta(T)^\gamma}$. Implied volatility is derived as $\sigma_{\text{SSVI}}(k,T) = \sqrt{w(k,T)/T}$.

14.3 Cross-Sectional Objective in Total Variance Space

At each minute t, we observe a set of market implied volatilities $\mathcal{D}_t = \{(K_i,\, \sigma_i^{\text{mkt}},\, F_i,\, T_i)\}_{i=1}^N$. We calibrate in total variance space to linearize the time dependency:

$$w_i^{\text{mkt}} = (\sigma_i^{\text{mkt}})^2 T_i, \qquad k_i = \log(K_i/F_i), \qquad T_i^{\text{safe}} = \max(T_i, \varepsilon_T)$$

14.3.1 Robust Loss Function (Huber Loss)

To make the fit robust to market microstructure noise and outliers (e.g., flash quotes in illiquid strikes), we employ a piecewise loss function known as the **Huber Loss** Huber [1964].

The residual $r_i(\mathbf{c}) = (w_{\text{SSVI}} - w_i^{\text{mkt}}) \times s$ is penalized as follows:

$$\ell(r) = \begin{cases} \frac{1}{2}r^2 & \text{if } |r| \leq \delta \quad \text{(Quadratic/Gaussian)} \\ \delta(|r| - \frac{1}{2}\delta) & \text{if } |r| > \delta \quad \text{(Linear/Laplacian)} \end{cases} \tag{36}$$

This ensures that small errors are minimized efficiently (Mean Squared Error behavior), while large outliers only contribute linearly, preventing them from dominating the gradient descent.

14.4 Expiry and ATM Weighting

The final objective function is a weighted average of the per-point losses:

$$\mathcal{L}_t(\mathbf{c}) = \frac{1}{W} \sum_{i=1}^{N} w_i \, \ell(r_i(\mathbf{c})), \quad \text{where } W = \sum_{i=1}^{N} w_i \tag{37}$$

The weights w_i are engineered to anchor the surface on the most information-dense regions of the volatility smile:

1. **Expiry Layer Weighting (w^{layer}):** We assign higher weights to short-dated expiries (e.g., $w \approx 3.0$ for $T^{(1)}$) and lower weights to long-dated expiries ($w \approx 1.0$), with linear interpolation. This prioritizes the accuracy of the front-month curve, which contains the most immediate market signal.

2. **ATM Band Up-Weighting (w^{ATM}):** Data points falling within a narrow log-moneyness band $|k_i| < 0.05$ receive a multiplicative boost (approx $1.5\times$). This enforces a tight fit around the ATM point, where liquidity and volume are highest Cont and Tankov [2004].

14.5 Initialization and Parameter Bounds

To ensure the optimizer converges to a global minimum rather than a local one, the SSVI fit is initialized using a deterministic two-step heuristic based on the current market cross-section. This reduces the search space for the gradient descent solver.

1. **ATM Total Variance Term Structure (c_0, c_1, c_2):** We first isolate the At-The-Money (ATM) subset of data where $|k_i| < 0.05$. On these points, we fit a low-order polynomial to the market total variances $\omega_i^{\text{mkt}} = (\sigma_i^{\text{mkt}})^2 T_i$:

$$\omega_i^{\text{mkt}} \approx c_1 T_i + c_2 T_i^2 \tag{38}$$

   The intercept c_0 is anchored to a small positive floor (≈ 0) to enforce the constraint that variance at $T = 0$ is zero. If data is insufficient, we revert to a conservative default vector and clamp the quadratic coefficient c_2 to prevent oscillating term structures Zeliade Systems [2012].

2. **Skew-Driven ρ Seed:** For the shortest expiry $T_{\min}$, we approximate the initial correlation ρ by performing a linear regression of IV σ^{mkt} against log-moneyness k. The resulting slope is linearly mapped and clipped to the domain $[-0.9, 0.5]$.

The remaining parameters are initialized to moderate defaults (e.g., $\gamma \approx 0.5$). The optimization is constrained within a tight axis-aligned box to prevent physical violations:

- ρ is bounded away from ± 1 to prevent extreme, unrealistic skew.
- γ is constrained to $[0.25, 0.5]$ to avoid excessively flat or steep smiles that violate the "moment formula" limits Gatheral [2012].
- The quadratic coefficients in $\theta(T)$ and $\lambda(T)$ are tightly bounded to prevent "explosive" variance behavior at long maturities.

14.6 Temporal Independence

The calibrations described above are performed *independently at each minute* for calls and puts. We deliberately exclude explicit time-series penalties or mean-reverting priors in the static objective. This design choice delegates all temporal smoothing responsibilities to the Extended Kalman Filter (EKF) described in Section 15. This separation of concerns allows us to evaluate the "raw" fit quality before applying temporal regularization.

14.7 Empirical Behavior

On the test sample (AMD, March 2025), this engineered SSVI specification demonstrated significant stability improvements over unconstrained models:

- **Robustness:** Delivers stable fits even when expiries and strikes are sparse or unevenly populated, a common scenario in pre-market or lunch-hour trading.
- **Profile Integrity:** Avoids the pathological "W-shaped" or oscillatory $\theta(T)$ profiles often produced by unconstrained polynomial fitting on noisy data.
- **Smoothness:** Produces minute-by-minute parameter paths that are sufficiently continuous, reducing the burden on the subsequent EKF smoothing step.

From an engineering perspective, the combination of robust Huber loss, expiry/ATM weighting, and tight box constraints achieves a practical balance between the theoretical rigor of SSVI and the noisy reality of American options microstructure Broadie and Detemple [1996].

15 Extended Kalman Filter (EKF)

15.1 Initial Hypothesis

After stabilizing the cross-sectional fits (engineered SABR and SSVI), we hypothesize that an *Extended Kalman Filter* (EKF) can further enhance the temporal consistency and forecasting quality of the surface parameters. This approach is grounded in three key assumptions:

- **Signal vs. Noise:** The independent, minute-by-minute cross-sectional calibrations are noisy snapshots of a latent, slowly evolving volatility state.
- **Bayesian Smoothing:** A Kalman-type filter functions as a recursive Bayesian estimator, optimally weighing the assumed state dynamics (prior) against the noisy IV observations (likelihood) Kalman [1960].
- **Non-Linearity:** Since the mapping from model parameters to Implied Volatility (IV) in both SABR and SSVI is highly non-linear, a standard Kalman Filter is insufficient. We require the EKF to linearize the observation manifold via Jacobian matrices Javaheri [2005].

15.2 Per-Regime State Representation

To avoid the computational complexity and potential crosstalk of a single high-dimensional filter, we implement a **Decoupled Filtering Architecture**. We maintain four independent filters:

1. SABR parameters for Calls ($\mathbf{x} \in \mathbb{R}^6$)
2. SABR parameters for Puts ($\mathbf{x} \in \mathbb{R}^6$)
3. SSVI parameters for Calls ($\mathbf{x} \in \mathbb{R}^8$)
4. SSVI parameters for Puts ($\mathbf{x} \in \mathbb{R}^8$)

For each regime, the state vector is initialized using the corresponding static cross-sectional fit from the first valid minute, with an initial covariance matrix $\mathbf{P}_0$ set to a scaled identity matrix.

15.3 Random-Walk State Dynamics

We model the temporal evolution of the parameters as a pure random walk (martingale prior). This choice is deliberate: it avoids imposing an arbitrary mean-reverting level that might bias the filter during regime shifts (e.g., an earnings announcement).

$$\mathbf{x}_t = f(\mathbf{x}_{t-1}) + \mathbf{w}_{t-1}, \qquad \text{where } f(\mathbf{x}) = \mathbf{x} \tag{39}$$

The process noise $\mathbf{w}_{t-1}$ is assumed to be Gaussian with covariance $\mathbf{Q} = q^2 I_d$, where q is a tunable hyperparameter representing the expected "volatility of the parameters" between time steps. Since the transition function f is identity, the transition Jacobian is trivial:

$$\mathbf{F}_t = \frac{\partial f}{\partial \mathbf{x}} = I_d$$

Consequently, the EKF **Prediction Step** reduces to simply inflating the state covariance matrix: $\mathbf{P}_{t|t-1} = \mathbf{P}_{t-1|t-1} + \mathbf{Q}$.

15.4 Non-Linear Observation Model

At each minute t, we observe a vector of market implied volatilities $\mathbf{z}_t = (\sigma_{t,1}^{\text{mkt}}, \dots, \sigma_{t,M}^{\text{mkt}})^\top$ corresponding to the available strikes and expiries. The relationship between the latent state and the measurement is non-linear:

$$\mathbf{z}_t = h(\mathbf{x}_t) + \mathbf{v}_t, \qquad \mathbf{v}_t \sim \mathcal{N}(0, \mathbf{R}) \tag{40}$$

Here, $h(\mathbf{x}_t)$ represents the pricing engine (SABR approximation or SSVI formula) that maps the parameter vector to theoretical IVs. The measurement noise covariance is isotropic, $\mathbf{R} = r^2 I_M$, implying equal confidence in all market quotes for this baseline implementation.

15.5 Numerical Jacobian Approximation

The core of the EKF **Update Step** requires the Jacobian matrix $\mathbf{H}_t$, which linearizes the observation function h around the predicted state:

$$\mathbf{H}_t = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\mathbf{x}_{t|t-1}} \in \mathbb{R}^{M \times d}$$

Deriving analytical gradients for the complex Hagan SABR approximation or the SSVI surface is error-prone and computationally cumbersome. Instead, we compute $\mathbf{H}_t$ using **Finite Difference Approximations** Pre [2007]:

$$\mathbf{H}_t^{(i,j)} \approx \frac{h_i(\mathbf{x} + \varepsilon e_j) - h_i(\mathbf{x})}{\varepsilon} \tag{41}$$

where e_j is the basis vector for the j-th parameter and ε is a small perturbation step. This matrix dictates how the "Innovation" (the difference between observed and predicted IV) is distributed across the state parameters during the Kalman Gain update.

15.6 EKF Recursion

With these ingredients defined, each regime follows the standard discrete-time Extended Kalman Filter recursion Welch and Bishop [2006].

Predict Step (Time Update) Project the state and covariance forward using the process model:

$$\hat{\mathbf{x}}_{t|t-1} = f(\hat{\mathbf{x}}_{t-1|t-1}) \tag{42}$$

$$\mathbf{P}_{t|t-1} = \mathbf{F}_t \mathbf{P}_{t-1|t-1} \mathbf{F}_t^\top + \mathbf{Q} \tag{43}$$

(Note: Since $f(\mathbf{x}) = \mathbf{x}$ and $\mathbf{F}_t = \mathbf{I}$, this simplifies to $\mathbf{P}_{t|t-1} = \mathbf{P}_{t-1|t-1} + \mathbf{Q}$.)

Update Step (Measurement Update) Correct the predicted state using the new observation vector $\mathbf{z}_t$:

$$\tilde{\mathbf{y}}_t = \mathbf{z}_t - h(\hat{\mathbf{x}}_{t|t-1}) \qquad \text{(Innovation)} \tag{44}$$

$$\mathbf{H}_t = \left.\frac{\partial h}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_{t|t-1}} \qquad \text{(Jacobian via Finite Diff.)} \tag{45}$$

$$\mathbf{S}_t = \mathbf{H}_t \mathbf{P}_{t|t-1} \mathbf{H}_t^\top + \mathbf{R} \qquad \text{(Innovation Covariance)} \tag{46}$$

$$\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{H}_t^\top \mathbf{S}_t^{-1} \qquad \text{(Optimal Kalman Gain)} \tag{47}$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + \mathbf{K}_t \tilde{\mathbf{y}}_t \qquad \text{(State Update)} \tag{48}$$

$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t \mathbf{H}_t) \mathbf{P}_{t|t-1} \qquad \text{(Covariance Update)} \tag{49}$$

15.6.1 Numerical Stabilization

To guard against numerical instability during matrix inversion, a small multiple of the identity ($\lambda \mathbf{I}$) is added to the innovation covariance $\mathbf{S}_t$ if its condition number exceeds a safety threshold Higham [1996]. Additionally, the update is skipped entirely if the dimensions of $h(\hat{\mathbf{x}}_{t|t-1})$ and $\mathbf{z}_t$ do not match (e.g., due to missing or inconsistent market data feeds).

15.7 Evaluation and Empirical Outcome

For each minute and regime, we evaluate the performance difference between:

- The **Static Parameters**: Obtained directly from independent cross-sectional calibration.
- The **EKF-Smoothed Parameters**: Obtained after applying the recursive filter.

In both cases, we compute the Model IVs against Market IVs on the same grid and log the per-minute Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

On the AMD test dataset, these diagnostics revealed the following insights:

1. **Loss Trade-off:** The EKF-smoothed parameters produce cross-sectional errors very similar to the static fits for most timestamps. However, in short-dated, high-volatility regions, the aggressive smoothing can slightly increase front-month RMSE by dampening genuine high-frequency market moves Simon [2006].

2. **Baseline Quality:** Because the static SSVI and SABR calibrations are already strongly regularized (via volume weighting, robust Huber loss, and box constraints), the raw parameter paths are not extremely noisy to begin with. Consequently, the incremental benefit of EKF smoothing is moderate rather than transformative.

3. **Model Misspecification Sensitivity:** The EKF is optimal only under correctly specified state dynamics and observation noise. In reality, the evolution of volatility parameters is richer than a Random Walk (e.g., mean reversion, jumps), and observation residuals are highly heteroskedastic across strikes and expiries. This structural mismatch limits the gains of a globally tuned EKF compared to the locally optimized static fits Maybeck [1979].

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